

Homework #2

Oscillators

The objective of these exercises is to study and understand the principle of sinusoidal oscillators from 2 opamp circuits: the Wien oscillator and the RC phase shifter oscillator.

It consists of 2 parts:

- part I : theoretical calculations.
- part II : LTSpice simulations.

Part 1 : Theory

Circuit in Fig. 1.1 is used to generate a sinusoidal signal. It consists of an amplifier with gain $A_0 = \frac{V_{S1}}{V_{E1}}$ (Figure 1.2), assumed to be a constant at first, and a reactive circuit with TF $\beta(j\omega) = \frac{V_{S2}}{V_{E2}}$ (Figure 1.3). D_1 and D_2 are silicon diodes with threshold $V_{D0} = 0,6$ V, zero series resistance when ON et and infinite when OFF. When switch K is open, these diodes make gain A_0 slightly nonlinear. When K is closed, the diode circuit is not active, having thus no effect on the oscillator's operation.

We set $R_1 = R = 100 \Omega$. C , R_2 et R_3 are unknown and need to be determined.

First, switch K is closed.

1. Give an expression for $\beta(j\omega)$.
2. What is the condition on A_0 and $\beta(j\omega)$ for the circuit to oscillate with a stable amplitude ?
3. Derive :
 - An expression for oscillation frequency F_0 .
 - The condition on R_2 et R_1 for a constant amplitude oscillation, without clipping.
4. Calculate values for C et R_2 so as to obtain a 50 kHz, constant amplitude oscillation.
5. What is its peak-to-peak amplitude. In practice, how can it be varied ?

Previous calculations assume that gain A_0 does not depend on frequency. Actually, the amplifier is a 1st order system, with gain G depending on frequency :

$$G(j\omega) = G(j2\pi f) = \frac{A_0}{1 + j \frac{f}{F_c}}$$

with F_c the intrinsic cutoff frequency of the opamp.

6. We want the oscillator to operate at frequency F_c , which is not *a priori* known. What are the values of gain and phase shift induced by the amplifier at this frequency?
7. Derive conditions on R , C and A_0 for the circuit to oscillate at constant amplitude, without clipping.
8. The opamp has a GBP of 3 MHz. Calculate values of F_c and C .

Now, switch K is open, and the oscillation frequency is the same as in question 4.

9. Plot the current vs. voltage characteristics of one of the diodes D_1 or D_2 .

In the 2 next questions, assume that D_1 and D_2 are OFF.

10. Give an expression for the gain of the amplifier.
11. Express voltage V_{R3} (Figure 1.2) vs. V_e .
12. Starting with D_1 and D_2 OFF, what is the condition on V_e for diode D_2 to become ON? What is, then, the state of D_1 ?
13. In that cas, redraw the equivalent circuit around the amplifier, and derive a relationship between V_s , V_e and V_{D0} whrn D_2 in ON and D_1 OFF.
14. Plot the transfer chacrcteristics $V_s = f(V_e)$ of the amplifier for $-15V < V_e < +15V$.
15. What is the purpose of diode D_1 and D_2 ? Please justify your answer.
16. What is the conditions on R_3 for the output oscillation to start?
17. Give an expression for output peak amplitude V_{smax} vs. V_{D0} , R_1 et R_2 .

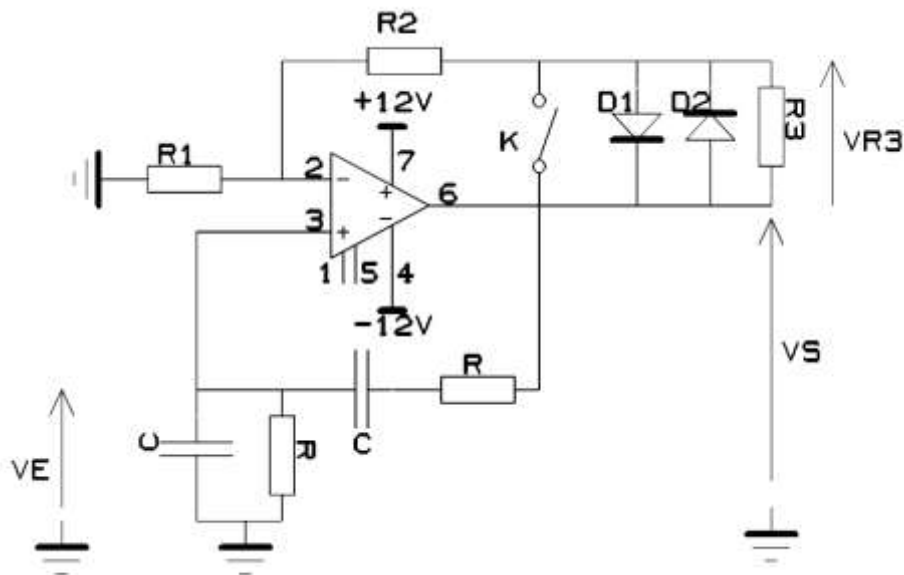


Figure 1.1 : full schematic of the oscillator.

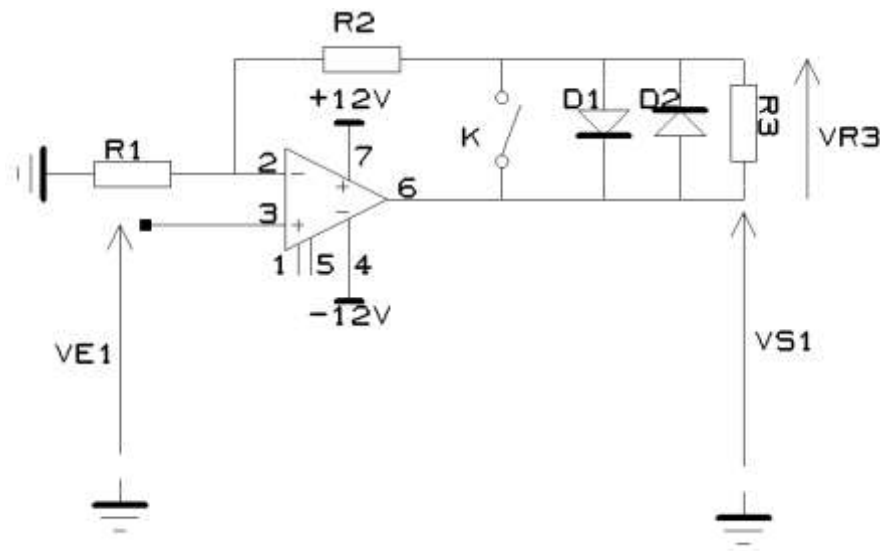


Figure 1.2 : amplifier.

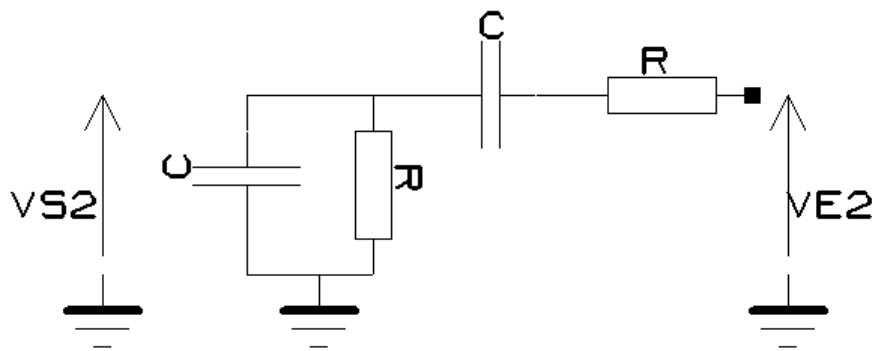


Figure 1.3 : reactive circuit.

Part 2 : Simulation

➤ Wien oscillator

Ce circuit is given in Fig. 2.1. Switch K is closed, so the diodes are not active.

1. Calculate the value of oscillation frequency F_0 in Hz and of resistor R_2 to maintain a stable oscillation.
2. Simulate the circuit using LTSpice, with UA741 opamp (see homework #1). Take R_2 slightly higher than theory. Simulate in 'Transient' mode, and choose a sufficient stop time to observe steady-state. To observe the oscillation startup, select « start external voltage supply at 0V »
3. Compare theory with simulation and conclude.
4. We know want the oscillator to operate at the cutoff frequency of the amplifier. Run a simulation to validate the results from question 6 in part I.

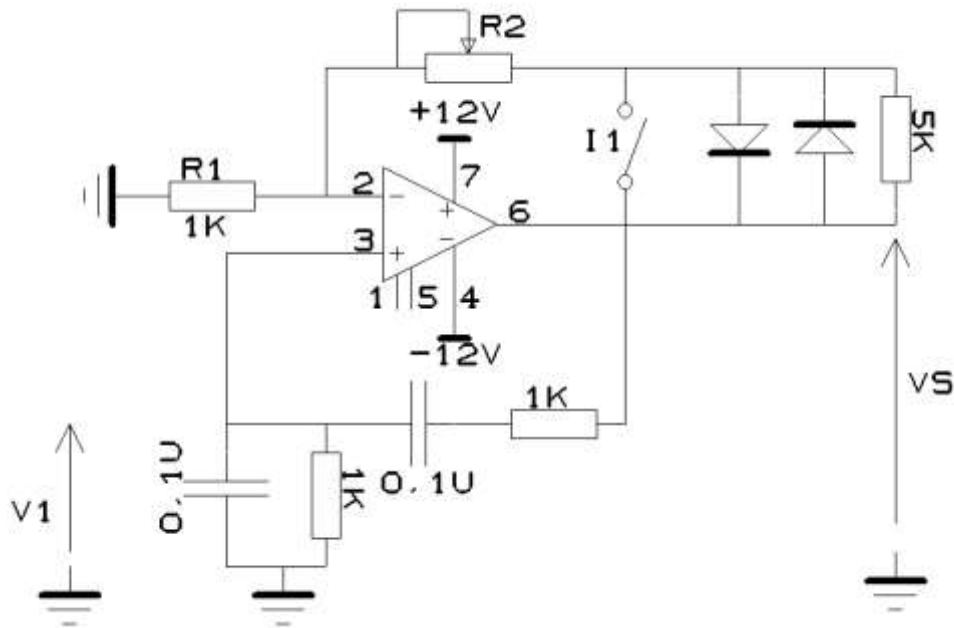


Figure 2.1 : Wien oscillator

➤ Oscillator with RC phase shifter

The circuit is give, in figure 2.2. Switch K is closed, so the diodes are not active.

1. Using LTSpice, simulate TF $\beta_{RC}(j\omega) = V_1/V_s$ (amplitude and phase) of the RC phase shifter alone, not connected to the amplifier.
- 👉 the RC phase shifter is loaded by R_1 (Fig. 2.2). which should be included in the circuit schematic.
1. Compare with theory (see slides of the course on Oscillators).
2. Derive frequency F_0 in Hz et the value of R_2 for stable oscillations.
3. Simulate the circuit of Fig. 2.2 using LTSpice, with UA741 opamp (see homework #1). Take R_2 slightly higher than theory. Simulate in 'Transient' mode, and choose a sufficient stop time to observe steady-state. To observe the oscillation startup, select « start external voltage supply at 0V »
4. Compare theory with simulation and conclude.

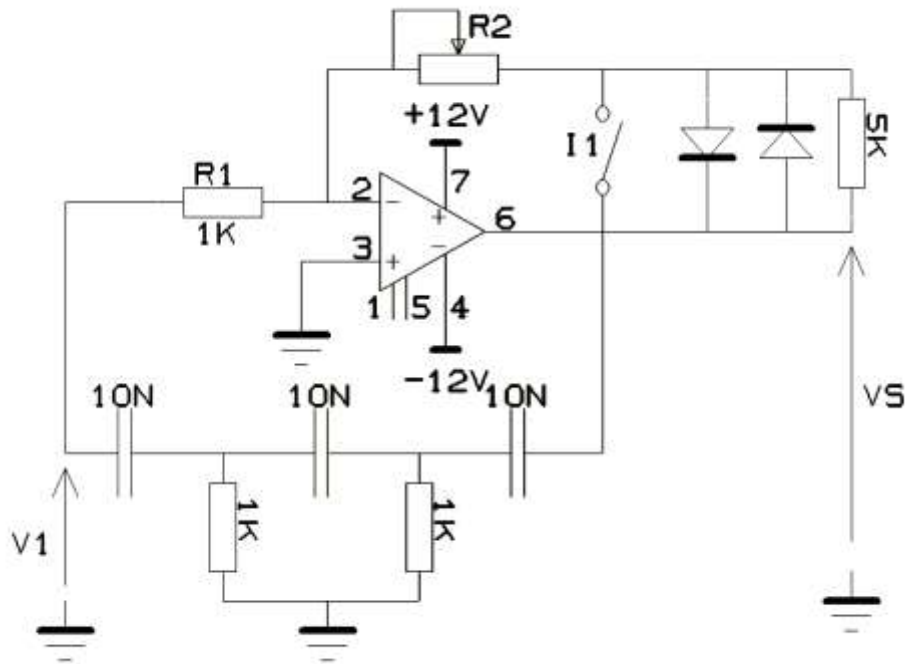


Figure 2.2 : oscillateur à déphaseur RC

➤ Amplitude control

Let's go back to the Wien oscillator (Fig. 2.1), but now switch I_1 is open. We want to study the effect of the diodes.

1. Using LTSpice, simulate the transfer characteristics V_S vs. V_I of the amplifier alone, including diodes (Fig. 2.3). For the purpose, set simulation mode to « DC sweep » corresponding to an input voltage ramp. For the diodes, use the standard model in the LTSpice library.
2. For $R_2 = 1,5 \text{ k}\Omega$, show how the amplitude of V_S can be predicted from transfer characteristics.
3. For $R_2 = 1,5 \text{ k}\Omega$, simulate the full Wien oscillator (Fig. 2.2), including diodes.
4. Compare amplitude of V_S obtained from simulation and from the previous question.
5. Compare with theory (question 17), if you have gone that far.

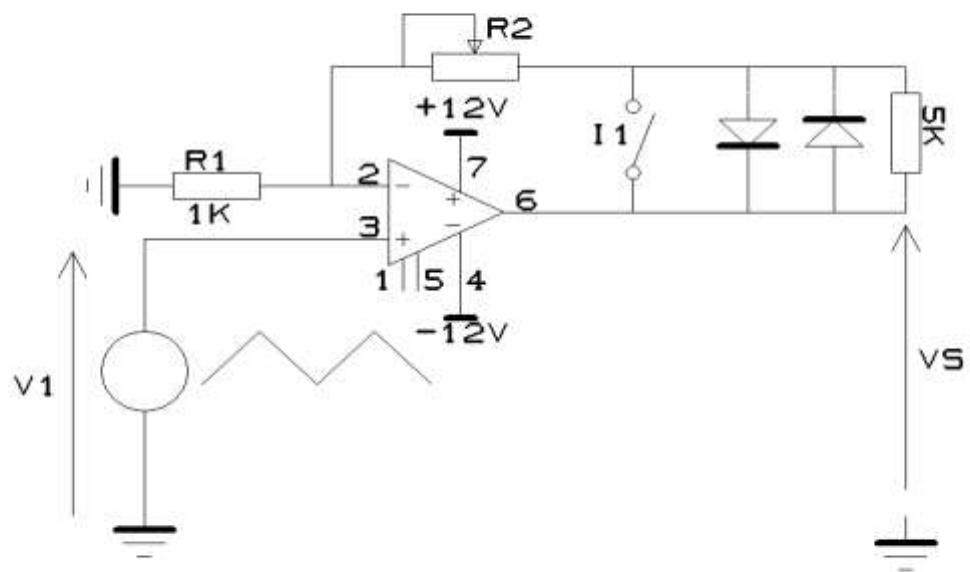


Figure 2.3 : nonlinear amplifier